Weak-Signal Ladder for Early AV Edge-Case Triage

A Minimal Event Contract and Promotion Logic

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Abstract

This paper proposes a compact, inspectable triage layer for autonomous vehicles that captures weak, short-lived edge-case findings and promotes them through a bounded state ladder (V1–V4, R0, Q0). Deliverables include a minimal event schema, deterministic promotion predicates with timing windows, planner action bands per state, and a minimal transition record for V3/V4 transitions. The module is intended as an insertion point between local anomaly emitters and the planner; it is not a perception module, a planner, or a certification framework. We provide example thresholds, pseudocode, and a reproducible evaluation plan to measure false escalation, missed early-warning, and planner stability.

1. Introduction

Many autonomous-driving failures are not failures of raw perception alone. They are failures of promotion control. A system sees something weak, partial, or locally inconsistent, but has no disciplined middle state between "ignore" and "treat as hazard." That gap matters in common edge-case conditions: partial occlusion, poor visibility, uncertain actor behavior, weak road-surface cues, or momentary disagreement across sensing routes.

This paper presents a minimal, inspectable triage module that: (1) represents weak local findings as bounded event objects, (2) promotes them through a short-lived state ladder under explicit timing and corroboration predicates, and (3) exposes auditable transitions and planner action bands for short-horizon caution.

The deliverable is not a theory of edge cases, but a minimal insertion-point contract between weak local anomaly emitters and bounded planner caution.

2. Transfer Claim and Scope

This paper does not propose a perception model, planner, controller, safety case, certification argument, vendor comparison, or deployment claim; it is scoped strictly to early triage: representation, promotion logic, planner action bands, and auditability.

3. Design Deliverable

3.1 State ladder

The ladder distinguishes informational states from action-constraining states.

State	Meaning	Class	Allowed downstream effect
V0	No active edge-case candidate	Informational	none
V1	Local weak finding recorded	Informational	attention weighting only
V2	Same candidate persists within short horizon	Informational	increased monitoring, request corroboration
V3	Candidate corroborated by second channel or repeated observation	Advisory	bounded planner caution allowed
V4	Candidate escalated under explicit predicates	Advisory / escalated	bounded speed/path conservatism allowed; audit required
R0	Candidate resolved back to corridor	Terminal	clear advisory state
Q0	Candidate quarantined as artifact / non-hazard explanation	Terminal	suppress further promotion unless re-opened

Corroboration counts only independent support. Multiple detections derived from the same underlying sensing route or tightly coupled processing stack do not count as separate corroborators unless independence has been explicitly validated.

State-class rule:

- **V1–V2 are informational only.** They may change short-horizon attention but may not directly alter committed route or persistent world state.
- **V3–V4 are advisory.** They may constrain planner choice within a declared narrow band.
- **R0/Q0 are terminal.** They close the candidate unless a new event ID is opened.

3.2 Minimal event schema

Each weak-signal candidate is represented by one bounded event object.

```
{
  "event_id": "string",
  "scene_id": "string",
  "timestamp_open_ms": 0,
  "timestamp_last_ms": 0,
  "state": "V1|V2|V3|V4|R0|Q0",
  "primary_type": "occlusion_actor|surface_state|actor_behavior|sensor_disagr
eement|control_irregularity|other",
  "source_channels": ["camera_front", "radar_front", "planner_trace"],
  "location_ref": {"lane_id": "string", "s_range_m": [0.0, 0.0]},
  "finding_score": 0.0,
  "confidence": 0.0,
  "persistence_ms": 0,
  "corroboration_count": 0,
  "supporting_features": ["bbox_edge_motion", "lateral_jerk_anomaly"],
  "planner_action_band": "none|monitor|soft_caution|bounded_caution",
  "audit_flags": ["short_horizon_only"],
  "closure_reason": "none|resolved|artifact|expired"
}
```

3.3 Field semantics

A minimal integration only needs the following semantics.

source_channels lists independent sensing or processing routes; corroboration_count is the count of independent channels or temporally separated confirmations that meet declared independence criteria.

All scores are normalized to [0,1] using the upstream module's calibration; triage should record the upstream calibration version in the event or transition record.

- `finding_score`: normalized weak-signal severity in [0,1]
- `confidence`: confidence in the event formation path, not hazard truth
- `persistence_ms`: time since first sighting under the current event id
- `corroboration_count`: independent support count across channels or repeated observation windows
- `planner_action_band`: the maximum action class allowed from this event

3.4 Timing windows and example thresholds

The thresholds below are a conservative reference profile for short-horizon urban scenarios; they are illustrative and must be validated per platform. Any evaluation should include a sensitivity analysis showing metric dependence on these thresholds.

- event open threshold: `finding_score >= 0.35`
- V1 → V2 persistence window: `>= 400 ms`
- V2 → V3 corroboration rule: `corroboration_count >= 2` within `<= 1500 ms`

- V3 → V4 escalation rule: `finding_score >= 0.60` and `confidence >= 0.55` and `persistence_ms >= 1000`
- event expiry to R0: no support for `1200 ms`
- event quarantine to Q0: explicit artifact evidence or contradictory evidence outweighs support by declared margin

3.5 Interface mapping to an AV stack

The triage layer sits between local anomaly emitters and stronger planner consequences.

Upstream module	Emits	Triage consumes	Triage emits	Downstream consumer
Perception	local weak finding	sensor-local anomalies	event open/update	prediction, planner
Prediction	actor trajectory irregularity	future-path uncertainty cue	event update	planner
Control / vehicle dynamics	slip / correction irregularity	control anomaly cue	event update	planner, safety monitor
Triage layer	event state	ladder logic	advisory band + transition record	planner, logger
Planner	candidate maneuver set	advisory band	constrained selection	controller

Triage is authoritative for event state; planners may read planner_action_band but must not modify event state.

3.6 Allowed planner actions by state

State	Planner allowance
V1	no route/path change; increase monitoring weight only
V2	no committed maneuver change; request corroborating evidence; optionally cap acceleration aggressiveness
V3	permit soft caution: modest speed reduction, increase following margin, defer lane change
V4	permit bounded caution: stronger speed reduction, temporary path conservatism, optional supervised review flag

State	Planner allowance
R0/Q0	clear advisory constraints

Weak signals do not directly control the vehicle; they constrain the planner's allowable action band in explicit, narrow ways. When the planner acts under a planner_action_band, it should log the chosen maneuver, the constraining event_id or event_ids, and the local rationale.

4. Transition Logic

4.1 Candidate pseudocode

```
def update_event(event, finding, now_ms):
    event.timestamp_last_ms = now_ms
    event.finding_score = max(event.finding_score, finding.score)
    event.confidence = max(event.confidence, finding.confidence)
    event.supporting_features.append(finding.feature_id)
    event.source_channels = union(event.source_channels, [finding.channel])
    event.corroboration_count = count_independent_support(event)
    event.persistence_ms = now_ms - event.timestamp_open_ms

    if artifact_evidence(event):
        event.state = "Q0"
        event.planner_action_band = "none"
        event.closure_reason = "artifact"
        return event

    if event.state == "V1" and event.persistence_ms >= 400:
        event.state = "V2"
        event.planner_action_band = "monitor"

    if event.state in ["V1", "V2"] and event.corroboration_count >= 2 and eve
nt.persistence_ms <= 1500:
        event.state = "V3"
        event.planner_action_band = "soft_caution"

    if event.state == "V3" and event.finding_score >= 0.60 and event.confiden
ce >= 0.55 and event.persistence_ms >= 1000:
        event.state = "V4"
        event.planner_action_band = "bounded_caution"

    if support_expired(event, now_ms, expiry_ms=1200):
        event.state = "R0"
        event.planner_action_band = "none"
        event.closure_reason = "resolved"

    return event
```

- Independence check: corroboration_count must exclude channels with known correlated failure modes.

- Staleness check: do not promote on corroboration if corroborating channel latency exceeds a configured bound.
- Channel fault rule: if a channel reports a self-diagnostic fault, ignore its contributions and mark the transition record with channel_fault=true.

4.2 Transition record

Every V3/V4 transition emits a compact transition record.

```
{
  "event_id": "string",
  "timestamp_ms": 0,
  "from_state": "V2",
  "to_state": "V3",
  "trigger": "corroboration_count>=2",
  "finding_score": 0.58,
  "confidence": 0.61,
  "source_channels": ["camera_front", "radar_front"],
  "planner_action_band": "soft_caution",
  "review_flag": false
}
```

Transition records should be append-only and stored with a monotonic sequence number and the triage module version.

5. Worked Example

5.1 Scenario: occluded actor emergence at an urban curb

t = 0.0 s

- camera_front detects weak edge motion at curb occlusion boundary
- camera confidence: 0.35
- radar_front: no return
- prediction stack flags lateral path uncertainty increase: +0.18 over local baseline

Event opened:

```
{
  "event_id": "EV_1042",
  "state": "V1",
  "primary_type": "occlusion_actor",
  "source_channels": ["camera_front", "prediction_trace"],
  "finding_score": 0.38,
  "confidence": 0.36,
  "persistence_ms": 0,
  "corroboration_count": 1,
  "planner_action_band": "none"
}
```

Planner effect: no change.

t = 0.4 s

- second camera frame maintains edge motion
- lateral uncertainty continues
- no radar confirmation yet

Event update:

- persistence_ms = 400
- state transitions V1 → V2
- planner_action_band = monitor

Planner effect: cap longitudinal acceleration increase for one horizon.

t = 1.0 s

- radar_front weak return appears
- camera confidence rises to 0.52
- predicted actor emergence probability increases

Event update:

```
{
  "event_id": "EV_1042",
  "state": "V3",
  "finding_score": 0.58,
  "confidence": 0.61,
  "persistence_ms": 1000,
  "corroboration_count": 2,
  "planner_action_band": "soft_caution"
}
```

Planner effect: reduce target speed by 10%, defer discretionary lane change, widen margin.

t = 1.4 s

- actor leg/torso partially resolves; camera confidence 0.69
- event meets V4 escalation rule

Event update:

- state transitions V3 → V4
- planner_action_band = bounded_caution

Planner effect: stronger speed reduction and temporary path conservatism.

5.2 Why this example matters

The example shows promotion under explicit timing and corroboration predicates with bounded planner effects at V3/V4.

6. Evaluation Plan and Metrics

A minimal evaluation compares a baseline AV stack without the weak-signal ladder to the same stack with the ladder enabled.

6.1 Primary metrics

- **False escalation rate**: fraction of V3/V4 promotions that later resolve as artifact or non-hazard
- **Missed early-warning rate**: fraction of true hazard cases where no V3/V4 state was reached before the hazard became explicit
- **Mean time-to-escalation**: time from first weak cue to V3 and V4
- **Planner stability score**: variance or jerk-related instability induced by advisory constraints relative to baseline
- **Quarantine precision**: fraction of Q0 outcomes later confirmed as artifact / non-hazard explanation

6.2 Experimental protocol

A minimal reproducible protocol is enough for this bridge paper. Use at least three scenario buckets: occluded actors, low-friction or degraded-surface cues, and unusual actor behavior. For each bucket, replay a fixed labeled scenario set through (1) a baseline stack without triage and (2) the same stack with triage enabled. Record V-state transitions, planner constraints, and final scenario outcomes. Compute false escalation rate, missed early-warning rate, mean time-to-escalation, and planner stability. Then sweep the three main thresholds — event-open threshold, V1 to V2 persistence threshold, and V2 to V3 corroboration window — to show sensitivity to calibration. The triage layer is useful only if it reduces missed early warning without unacceptable planner instability or excessive false escalation.

7. Governance and Audit Primitives

Governance here means a narrow engineering boundary: versioned transition predicates, append-only transition records for V3/V4 promotions, and optional review flags for supervised testing. It does not mean organizational governance, regulatory compliance, or production deployment policy.

Minimum engineering artifact set:

- event object
- transition record

- state-to-action-band table
- review flag for supervised evaluation contexts

These primitives make promotion inspectable in testing and evaluation contexts; production governance requires additional organizational and security controls beyond this paper.

8. Limits and Non-Goals

This triage module intentionally does not solve:

- hazard classification in general
- end-to-end planning
- AV safety certification
- controller design
- full logging infrastructure
- production threshold calibration

These exclusions keep the triage layer narrowly scoped to inspectable promotion logic.

9. Conclusion and Next Step

This paper defines a compact triage module: a state ladder, a minimal event schema, deterministic promotion predicates, planner action bands, and a minimal transition record for early AV edge-case triage.

A separate downstream disagreement-binding layer such as SDB-1 could consume estimator conflict across perception, planning, or reflex-like modules and trigger reversible degradation when coherence fails, but that function is outside the scope of the present triage module.

Implement the triage module as a deterministic, versioned component; run the reproducible experiment described in Section 6; and publish the scenario traces and audit logs. That will show whether the ladder reduces missed early warnings without unacceptable planner instability.